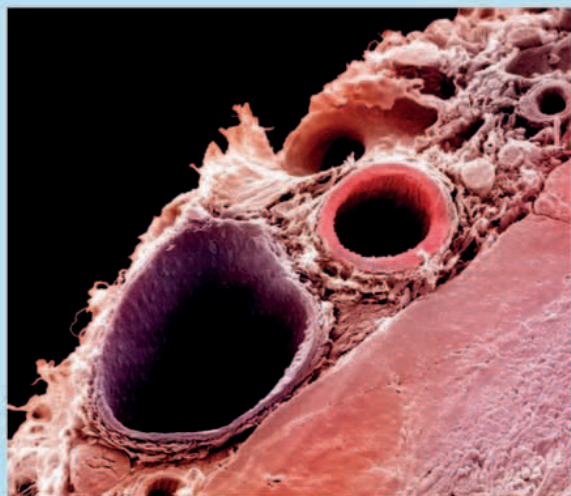




LOOK CLOSER: SIDE BY SIDE

Arteries and veins run side by side throughout your body, forming an amazingly complex network of tubes. These blood vessels divide continuously, so that they can reach every part of your body.



▲ **ARTERIES AND VEINS** This cross-section of an organ shows a thin-walled vein (purple) alongside a thicker-walled artery (red). They are held in place and supported by connective tissue.

CAPILLARY NETWORK

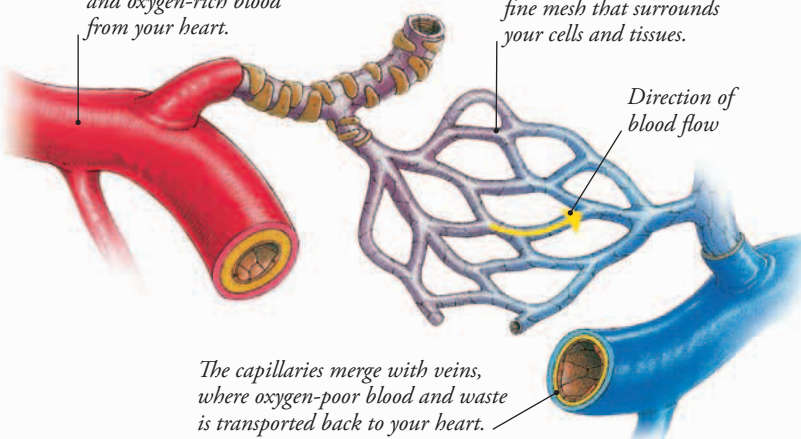
Your arteries and veins are the main highways of the circulatory system. Your capillaries are like the side streets and back alleys that link those highways. As the arteries get closer to their destination, they get smaller and form a capillary network. This ensures that every cell gets supplies of food and oxygen from a capillary. Once their deliveries are made, capillaries merge again to form tiny veins.

The arteries carry nutrients and oxygen-rich blood from your heart.

The capillaries form a fine mesh that surrounds your cells and tissues.

Direction of blood flow

The capillaries merge with veins, where oxygen-poor blood and waste is transported back to your heart.



WOUND HEALING

When a cut in the skin damages a blood vessel, the body reacts to prevent infection and stop bleeding. As white blood cells move in to destroy invading germs, platelets form a plug to stop blood from leaking out. They also create a net of fibers that trap blood cells to make a jellylike clot. This dries to form a hard scab, which protects the wound while it heals.

Blood leaking from a small cut takes about five minutes to clot.

